Self-Adaptive Improved Fungal Growth Optimization: A Hybrid Metaheuristic Framework for High-Quality, Robust and Scalable Clustering

Review comments received on 24-June-2026
Submission deadline: 22- August-2026
Journal: Knowledge and Information Systems (KAIS)
Submission ID: 13936815-b829-4668-a577-830684410d13

Author's responses to the review comments

Reviewer 1

1. The abstract should clearly explain what problem is addressed and why it is important.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated in abstract section.

2. The abstract should briefly mention how the problem is solved and summarize the key results.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated in abstract section.

3. The motivation of the work should be clearly established in the introduction.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated in introduction, section 1 paragraph 3.

4. The introduction should clearly identify the research gap and how the proposed work addresses it.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated Introduction, section 1 paragraph 4.

5. The main contributions should be explicitly stated and clearly differentiated from existing methods.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated Introduction, section 1 paragraph 5.

6. The related work section should be comprehensive and include recent and relevant studies.
[Author's response]: The task has been executed in accordance with the review comment. Recent papers are added in the Recent Work, section 2.

7. The proposed methodology should be explained clearly with sufficient technical detail.
[Author's response]: The task has been executed in accordance with the review comment. Section 3.1, Section 3.3, Section 3.4 and Section 3.6 are updated.

8. A convergence analysis should be provided to justify the stability and reliability of the proposed approach.
[Author's response]: The task has been executed in accordance with the review comment. Section 5.3.1 is added.

9. A computational complexity analysis should be included and clearly defined.
[Author's response]: The task has been executed in accordance with the review comment. Section 3.5 is updated.

10. All algorithm parameters should be clearly described and properly justified.
[Author's response]: The task has been executed in accordance with the review comment. Table 2 under section 4 (Experimental Setup) is added to describe the algorithmic parameters.

11. The experimental setup should be described in sufficient detail to ensure reproducibility.
[Author's response]: The task has been executed in accordance with the review comment. Section 4 (Experimental Setup) is added to incorporate your comment.

12. Appropriate statistical analyses (e.g., Wilcoxon, Friedman, t-test) should be conducted to validate performance claims.
[Author's response]: The task has been executed in accordance with the review comment. Statical Analysis (Section-5.2) section is updated and table 13 is added to ensure the unified parameter configuration.

13. Results should be reported as mean and standard deviation over multiple independent runs.
[Author's response]: The task has been executed in accordance with the review comment. Table 4 to Table 12 is updated as per review comment.

14. An ablation study should be performed to demonstrate the contribution of each component of the proposed method.
[Author's response]: The task has been executed in accordance with the review comment. Section 5.3.2 is added as per your comment.

15. Comparisons should be made with relevant state-of-the-art methods under identical experimental conditions.
[Author's response]: The task has been executed in accordance with the review comment. Algorithms like SOA and SO are introduced (in subsection 3.6, Table-1).

16. Evaluation metrics should be clearly defined and appropriate for the problem domain.

[Author's response]: The task has been executed in accordance with the review comment. "Section 4.3 - Evaluation Matrices" is introduced.

17. Convergence behavior and performance trends should be clearly presented and discussed.
[Author's response]: The task has been executed in accordance with the review comment. Convergence Section 5.3.1. is introduced.

18. Limitations of the proposed approach should be clearly acknowledged and discussed.
[Author's response]: The task has been executed in accordance with the review comment. Limitations are introduced in Section 7.

19. The conclusion should be consistent with the experimental findings and avoid over-claiming.
[Author's response]: The task has been executed in accordance with the review comment. Conclusion (Section 9) is updated.

20. Future research directions should be realistic, technically sound, and well justified.
[Author's response]: The task has been executed in accordance with the review comment. Section 8 is introduced as Future Research.

21. It is recommended that the authors critically review and discuss the following work to position their study more clearly within the existing literature:
https://doi.org/10.1007/s44196-026-01363-3
https://doi.org/10.1038/s41598-025-09326-y
https://doi.org/10.1007/s11227-025-07810-w
https://doi.org/10.1109/ISED67359.2025.11405178
[Author's response]: The task has been executed in accordance with the review comment. All the recommended works are included under the Recent Work, Section 2.

Reviewer 2

1. Restructuring of the "Introduction" is required to provide a thorough overview of the study's context and motivation. This section should critically evaluate prior research limitations and clearly show how each contribution of this work fills those gaps. The current version of the manuscript does not adequately engage with the relevant literature, lacks a clear rationale for the research, and contains redundant descriptions.
[Author's response]: The task has been executed in accordance with the review comment. Incorporated in introduction, section 1.

2. The authors' rationale for selecting the FGO over other swarm intelligence algorithms should be substantiated with comparative evidence. Additionally, a more comprehensive description of how the optimization algorithms are enhanced to handle machine learning models is required. For instance, the literature review should be expanded to highlight existing research on optimizing the machine learning models with metaheuristic algorithms (e.g., doi.org/10.1016/j.asoc.2022.108766, doi.org/10.1016/j.eswa.2023.122349).

[Author's response]: The task has been executed in accordance with the review comment. All the recommended works are included under the Recent Work, Section 2.

3. The experimental design and results presented are insufficient to comprehensively validate the performance of the proposed algorithm. This section requires significant enhancement through ablation studies to isolate the contribution of key algorithmic components.
[Author's response]: The task has been executed in accordance with the review comment. Section 5.3.2. contains the Ablation Study of the proposed algorithm.

4. To assess computational efficiency, the runtime of each competing algorithm in the experiment should be recorded and compared.
[Author's response]: The task has been executed in accordance with the review comment. Table 4 to Table 12 contains the run time of each algorithm.

5. It is recommended to upload the software source code and the datasets used in the experiment to a public platform, such as GitHub, and provide clear instructions to enable other researchers to replicate the experiment and make further improvements.
[Author's response]: The task has been executed in accordance with the review comment. Source Code and Data Availability section is updated. The repository contains readme.txt as a description of the folders and files. The dataset.txt contains the google-drive link of datasets.

6. The overall academic quality of the English expression throughout the article is insufficient. It is therefore advisable to read the entire manuscript thoroughly and make careful revisions. It should be noted that even the abstract contains unclear expressions.
[Author's response]: We appreciate the reviewer's constructive comment regarding the language quality. We have taken this advice seriously and performed a comprehensive, line-by-line revision of the entire manuscript to improve academic expression, sentence structure and overall flow. Attention was paid to the abstract to eliminate any ambiguous phrasing. We hope the revised manuscript now meets the high standards of the journal.

Reviewer 3

The paper is interesting but following comments must be addressed:

1. The claimed novelty of SAIFGO is primarily based on combining existing techniques such as chaotic initialization, Lévy flights, opposition-based learning, adaptive parameter control, K-Means refinement, and FCM refinement. The paper does not clearly demonstrate which component contributes genuinely new methodological innovation beyond existing hybrid metaheuristic literature.
[Author's response]: The task has been executed in accordance with the review comment.

The true methodological innovation of SAIFGO lies in the synergistic integration and unique mathematical interplay of these components, which yields significantly superior balancing of exploration and exploitation compared to existing hybrids. As detailed in Section 5.3.2 (Ablation Study), we systematically removed and evaluated each core mechanisms.

2. The paper lacks a comprehensive ablation study. Since multiple enhancements are introduced simultaneously, it is impossible to determine the individual contribution of chaotic initialization, Lévy mutation, opposition-based learning, adaptive search, and hybrid refinement.
[Author's response]: The task has been executed in accordance with the review comment. An ablation study introduces in section 5.3.2.

3. The description of the self-adaptive mechanism is insufficient. The paper repeatedly claims a success-based adaptive strategy selection framework, but the mathematical formulation and implementation details of this mechanism are not provided in the algorithm or equations.
[Author's response]: The task has been executed in accordance with the review comment. Mathematical Formulation Section – 3.1, Algorithm Section 3.3 are updated.

4. There is a discrepancy between the algorithm description and the novelty section. The novelty section mentions elite-guided search, differential exploration, and adaptive strategy probabilities, but these operators do not explicitly appear in Algorithm 1, creating concerns about consistency and reproducibility.
[Author's response]: The task has been executed in accordance with the review comment. Algorithm Section – 3.3 and 3.4 are updated.

5. The composite fitness function uses equal weighting for all clustering validity indices without justification. Different indices have different scales and statistical properties, and equal weighting may introduce bias or dominance by certain metrics.
[Author's response]: The task has been executed in accordance with the review comment. All the indices are normalized and then used in the above equation to avoid bias or dominance by certain metrics.

6. The choice of datasets is relatively limited and does not adequately evaluate scalability. Most datasets contain only a small number of features, and there are no experiments on genuinely high-dimensional clustering problems despite claims of scalability.
[Author's response]: The task has been executed in accordance with the review comment. High Dim Datascape and Yeast Dataset are introduced which are high-dimensional datasets.

7. The paper claims superiority on high-dimensional problems, yet no experiments are reported on benchmark high-dimensional datasets where dimensionality exceeds hundreds or thousands of features.
[Author's response]: During our experimental design, we found a distinct lack of standard, publicly available benchmark datasets of this scale that fit the specific evaluation criteria of our study. To bridge

this gap in the literature and directly address your concern, we have curated and introduced two new, strictly high-dimensional datasets in the revised manuscript:

1. **Yeast Transcriptomics & Bioinformatics:** A highly complex biological dataset consisting of 6,071 samples (genes/variables) and 93 features.
2. **High-Dim Datascape:** A synthetic/curated dataset containing 537 features and 230 samples, specifically designed to stress-test algorithms in multi-dimensional spaces.

8. Several competing algorithms show nearly identical results across multiple datasets, sometimes matching to many decimal places. This raises concerns regarding implementation fairness, parameter tuning, or whether the competing methods were allowed sufficient optimization opportunities.
[Author's response]: The task has been executed in accordance with the review comment. Result section has been updated, and standard deviation is also added. Section 4.2 is introduced as Parameter Configuration. Ablation study section 5.3.2 has been added to showcase individual contributions.

9. Runtime analysis is missing. Although computational complexity is discussed theoretically, there are no empirical execution time comparisons, memory usage analyses, or scalability studies to support practical applicability.
[Author's response]: The task has been executed in accordance with the review comment. Execution time is shown from Table 4 – Table 12.

10. The statistical evaluation focuses heavily on cluster validity metrics but lacks confidence intervals, standard deviations, or repeated-run statistics. Since metaheuristics are stochastic, reporting only average performance is insufficient.
[Author's response]: The task has been executed in accordance with the review comment. Table 4 – Table 12 is introduced as a standard deviation format (datasets were run 30 times).

11. The paper presents Friedman, Nemenyi, and t-tests, but the experimental protocol is not described in sufficient detail. The number of independent runs, significance thresholds, multiple comparison corrections, and assumptions behind the tests are not clearly specified.
[Author's response]: The task has been executed in accordance with the review comment. Table 13 is introduced to showcase Unified Parameter Configuration Matrix for Statistical Validation.

12. The comparison excludes many recent state-of-the-art clustering optimizers and modern clustering approaches, including deep clustering methods, representation-learning-based clustering, and recent swarm intelligence algorithms, limiting the strength of the experimental conclusions.
[Author's response]: The task has been executed in accordance with the review comment. SOA, SO algorithm is introduced to compare the clustering validity indices comparison.

13. The proposed algorithm requires multiple clustering validity indices, K-Means refinement, FCM refinement, chaotic initialization, and adaptive updates, which significantly increase algorithmic complexity. The paper does not analyze whether the performance gain justifies this additional complexity.

[Author's response]: The task has been executed in accordance with the review comment. The Computational Complexity Analysis section-3.5 is updated.

14. Reproducibility is limited because important implementation details are missing, including parameter settings for Lévy flights, adaptive coefficient values, stopping criteria sensitivity, initialization seeds, and hyperparameter tuning procedures.
[Author's response]: The task has been executed in accordance with the review comment. For reproductivity concerns Hyperparameter Configurations (Table 2) is added in section 4.2.

15. The paper concludes that SAIFGO is a powerful alternative for complex clustering optimization problems, but the evidence is restricted to a small set of benchmark datasets. Broader validation on real-world large-scale clustering applications would be necessary before making such a strong general claim.
[Author's response]: The task has been executed in accordance with the review comment. Two high dimensional datasets (**Yeast Transcriptomics & Bioinformatics and High-Dim Datascape**) are introduced along with two practical applications.

Reviewer 4

This paper proposes SAIFGO, a clustering-oriented metaheuristic that stacks chaotic initialization, Lévy flight mutation, opposition-based learning, and a linear decay coefficient onto the recently published Fungal Growth Optimization (FGO) algorithm. Each of these four components is a well-established technique in the metaheuristic literature, and their straightforward combination does not constitute a substantive novel contribution. Collectively, the contradiction between claimed mechanisms and actual formulas, the statistically impossible replication of results across unrelated algorithms, and the copy-paste identity of results across different datasets render this manuscript unsuitable for publication in its current form.
[Author's response]: The task has been executed in accordance with the review comment.
The true methodological innovation of SAIFGO lies in the synergistic integration and unique mathematical interplay of these components, which yields significantly superior balancing of exploration and exploitation compared to existing hybrids. As detailed in Section 5.3.2 (Ablation Study), we systematically removed and evaluated each core mechanisms.

1. The claims in Section 3.4 (dynamic strategy selection, success-based adaptation, elite-mean guidance, reinforcement-learning-like feedback) are entirely unsupported by the mathematical formulation in Section 3.2, which contains only a standard PSO-style position update toward the global best (Eq. 17), a generic Lévy perturbation (Eq. 18), and basic OBL (Eq. 19).
[Author's response]: The task has been executed in accordance with the review comment. Mathematical Formulation (Section – 3.1) is updated.

2. The composite fitness function (Eq. 7) assigns equal unit weights to five validity indices with vastly

different scales (CH-Score reaches ~18000 while Dunn Index remains ~0.005), making the objective effectively dominated by a single term.
[Author's response]: The task has been executed in accordance with the review comment. All the indices are normalized and then used in the above equation to avoid bias or dominance by certain metrics.

3. Algorithm 1 contains unreadable corrupted text on Line 14 and omits critical implementation details including the OBL triggering condition and the local refinement schedule.
[Author's response]: The task has been executed in accordance with the review comment. Section 4.2 - Parameter Configurations is added.

4. Across Tables 4–10, all non-SAIFGO FCM-based hybrids (GA+FCM, PSO+FCM, GTO+FCM, BO+FCM, ISGO+FCM, EO+FCM, HO+FCM) produce identical validity index values to six decimal places on every dataset, which is statistically impossible for algorithms from fundamentally different paradigms with stochastic initialization.
[Author's response]: The task has been executed in accordance with the review comment. We have introduced the Standard Deviation format.

5. The SAIFGO+K-Means row in Table 8 (Placement Dataset) is byte-for-byte identical to the SAIFGO+K-Means row in Table 9 (Shopping Dataset) across all five indices, despite these being entirely different datasets.
[Author's response]: The task has been executed in accordance with the review comment. We have introduced the Standard Deviation format.

6. No ablation study was conducted, making it impossible to determine which, if any, of the four added components actually contribute to the reported improvements.
[Author's response]: The task has been executed in accordance with the review comment. Section 5.3.2 Ablation Study is added.

7. The baseline algorithms (GA, PSO) were introduced in 1975 and 1995 respectively, and the paper omits comparisons with recent state-of-the-art clustering optimizers.
[Author's response]: The task has been executed in accordance with the review comment. New algorithms like SOA, SO is introduced which are quite new optimization algorithms.

8. Thousands of statistical comparisons are performed across seven datasets, five indices, and twenty algorithm variants without any multiple-comparison correction.
[Author's response]: The task has been executed in accordance with the review comment. The clustering validity indices tables are introduced as Standard Deviation format.

9. Convergence analysis is restricted to the Iris dataset (150 samples), the smallest and least challenging benchmark in the study, and reports iteration count rather than wall-clock time.

[Author's response]: The task has been executed in accordance with the review comment. Convergence analysis has been done with a moderately high-dimensional dataset (Delivery Dataset) in section 5.3.1 Convergence Analysis.

Reviewer 5

1. The novelty of the proposed SAIFGO framework is limited because it mainly combines existing techniques such as chaotic initialization (Eqs. 14–15), Lévy flight mutation (Eq. 18), opposition-based learning (Eq. 19), and adaptive parameter control (Eq. 16), all of which are well-established strategies in the metaheuristic optimization literature. The work appears to be an incremental variant of FGO rather than a fundamentally new optimization framework.

[Author's response]: The task has been executed in accordance with the review comment.

The true methodological innovation of SAIFGO lies in the synergistic integration and unique mathematical interplay of these components, which yields significantly superior balancing of exploration and exploitation compared to existing hybrids. This is discussed in detail in section 3.4 (Novel contribution of SAIFGO). Also in Section 5.3.2 (Ablation Study), we systematically removed and evaluated each core mechanisms.

2. There is a noticeable inconsistency between the claimed contributions in Section 3.4 (pp. 9–10) and the actual algorithmic formulation. Several claimed mechanisms, such as elite-mean guidance, strategy competition, feedback-driven learning, and self-adaptive operator selection, are not clearly reflected in Algorithm 1 or supported by corresponding mathematical formulations.

[Author's response]: The task has been executed in accordance with the review comment. Section 3.3 - Algorithmic Blueprint is updated.

3. The manuscript does not include an ablation study to quantify the individual contributions of chaotic initialization, Lévy flight mutation, opposition-based learning, adaptive search control, and hybrid refinement. Therefore, it is difficult to determine which component is primarily responsible for the reported performance improvements.

[Author's response]: The task has been executed in accordance with the review comment. An ablation study is introduced under section 5.3.2.

4.The experimental results reported in Tables 4–10 (pp. 12–18) show many competing algorithms producing identical or nearly identical results across multiple datasets and metrics. Such behavior is unusual for stochastic metaheuristic algorithms and raises concerns regarding the experimental setup, implementation, or result reporting.

[Author's response]: The task has been executed in accordance with the review comment. Tables are introduced with standard deviation form.

5.Encourage the authors to discuss new and recent optimization algorithms, such as "Secant

Optimization Algorithm for efficient global optimization", and "Schrödinger optimizer: A quantum duality-driven metaheuristic for stochastic optimization and engineering challenges".
[Author's response]: The task has been executed in accordance with the review comment. Both SOA and SO are introduced.

6. Although several statistical analyses are presented in Section 4.2 (pp. 19–24), the manuscript does not provide rigorous statistical comparisons of SAIFGO against competing optimization algorithms over multiple independent runs. Consequently, the superiority claims are not sufficiently validated.
[Author's response]: The task has been executed in accordance with the review comment. The Statical Analysis section is updated, "Table 13 - Unified Parameter Configuration Matrix for Statistical Validation" is added.

7. The comparison study does not include many recent state-of-the-art clustering optimization methods, making it difficult to assess the true competitiveness of the proposed approach within the current literature.
[Author's response]: The task has been executed in accordance with the review comment. Recent state of the art methods like SOA and SA is added.

8. The computational complexity analysis on pp. 10–11 is oversimplified and does not fully account for the additional costs introduced by Lévy flight operations, opposition-based learning, validity-index evaluation, K-Means refinement, and FCM refinement.
[Author's response]: The task has been executed in accordance with the review comment. The computational complexity analysis is updated (section 3.5).

9. The manuscript lacks runtime analysis, convergence time comparison, and scalability experiments on large-scale or high-dimensional datasets. As a result, the practical computational efficiency of SAIFGO remains unclear.
[Author's response]: The task has been executed in accordance with the review comment. Execution time has been added in all results tables (table 4 -12). Convergence time comparison (section 5.3.1) is also discussed. Two new high-dimensional datasets are used in the results section (Yeast Transcriptomics & Bioinformatics and High-Dim Datascape)

10. The real-world application section (Section 5, pp. 24–27) is largely descriptive and does not provide quantitative performance evaluation, benchmark comparisons, or application-specific metrics to demonstrate the practical effectiveness of the proposed method.
[Author's response]: The task has been executed in accordance with the review comment. Validity indices of comparison (with FGO) for both real-life applications are added (table 16, 18 respectively).

11. One of the major limitations of traditional clustering methods discussed in the Introduction (p. 2) is the requirement to predefine the number of clusters. However, the proposed framework still relies on a fixed number of clusters and does not provide any mechanism for automatic cluster-number determination.

[Author's response]: The task has been executed in accordance with the review comment. The optimal number of clusters for each dataset has been determined using binary search and cluster validity indices [reference 41].

12. The paper provides limited theoretical support for the proposed optimizer. No convergence proof, theoretical stability analysis, or formal justification is presented to explain why the combination of the proposed operators should consistently improve optimization performance.

[Author's response]: The task has been executed in accordance with the review comment. Under 5.2 Section table 13 is introduced. Also convergence analysis (section 5.3.1), Statistical analysis (section 5.2) are given to substantiate the stability claim of the proposed algorithm.

13. The composite fitness function defined in Eq. (7) assigns equal importance to all clustering validity indices without justification or sensitivity analysis. The impact of this design choice on clustering quality is therefore unclear.

[Author's response]: The task has been executed in accordance with the review comment. All the indices are normalized and then used in the above equation to avoid bias or dominance by certain metrics.

14. I notice that many references that are related to the research topic are somehow old. Please make the reference robust with a recent close paper such as :"Large scale salp-based grey wolf optimization for feature selection and global optimization"; "An innovative quadratic interpolation salp swarm-based local escape operator for large-scale global optimization problems and feature selection"; "Enhancing feature selection with GMSMFO: A global optimization algorithm for machine learning with application to intrusion detection"; "An innovative time-varying particle swarm-based Salp algorithm for intrusion detection system and large-scale global optimization problems"; "An optimized computational framework for non-small cell lung cancer subtype classification and biomarker discovery";

[Author's response]: The task has been executed in accordance with the review comment. All the references are added under Recent-Work section.